INCH-POUND

MIL-A-47315A (MI)
27 September 1991
SUPERSEDING
MIL-A-47315 (MI)
8 May 1987

MILITARY SPECIFICATION

ADHESIVE, POLYURETHANE

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of two-part polyurethane adhesive.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8040

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

MMM-A-132	-	Adhesives, Heat Resistant, Airframe Structural, Metal-to-Metal
PPP-C-2020	-	Chemicals, Liquid, Dry, and Paste: Packaging of

STANDARDS

FEDERAL

FED-STD-175	-	Adhesives, Methods of Testing

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking
MIL-STD-45662	-	Calibration Systems Requirements

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D150	-	AC Loss Characteristics and Permittivity (Dielectric Constant) of Solid Electrical Insulating Materials, Standard Test Method for
ASTM D257	-	D-C Resistance or Conductance of Insulating Materials, Standard Test Methods for
ASTM D792	-	Specific Gravity (Relative Density) and Density of Plastics By Displacement, Standard Test Methods for
ASTM D2240	-	Rubber, Property-Durometer Hardness, Standard Test Method for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article inspection. When specified in the contract or procurement document, a first article sample shall be provided which shall meet the requirements of this specification (see 4.3 and 6.2).

3.2 Properties.

3.2.1 Uncured properties. The adhesive, as received, shall be polyurethane polyether base reversion resistant premixed and frozen. The container size (ounces, pints, quarts, or gallons) shall be determined by the contracting officer (see 6.2).

3.2.1.1 Specific gravity. The specific gravity shall be 1.25 $\pm$ 0.05 at a temperature of 25° $\pm$ 2°C, (see 4.6.2).

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3.2.1.2 Working life. The usable life of the mixed compound, exposed in an open container at 25° ± 2°C, shall be not less than 45 minutes (see 4.6.3).

3.2.2 Cured properties. The compound shall exhibit the following properties after curing for not less than 3 hours at a temperature of 71°C ± 3°C and allowed to cool to a temperature of 25° ± 2°C (see 4.6.4).

3.2.2.1 Overlap shear strength. (see 4.6.4.1).

Temperature	Shear strength, pounds/force per square inch (lbf/in^2)
25° ± 2°C	2000 ± 500
-54° ± 3°C	4000 ± 1000
68.5° ± 3°C	1000 ± 250

3.2.2.2 Hardness, Shore A. The Shore A hardness shall be 90 ± 5 (see 4.6.4.2).

3.2.2.3 Dielectric constant. The dielectric constant shall be not greater than 6.6, at 1 kilohertz (KHz) (see 4.6.4.3).

3.2.2.4 Volume resistivity. The volume resistivity shall be no less than 9×10^9 ohm/centimeter, at 25°C (see 4.6.4.4).

3.2.2.5 Copper corrosion. The compound shall not exhibit corrosive effects from copper (see 4.6.4.5).

3.2.2.6 Fungus resistance. The compound shall not exhibit effects to fungus nutrients (see 4.6.4.6).

3.2.2.7 Hydrolytic stability. The compound shall not exhibit a reversion characteristic. The loss of hardness shall be not greater than 10 (see 4.6.4.4).

3.3 Environmental test requirements.

3.3.1 Humidity. The cured product shall be subjected to a humidity test in accordance with method 507.3, MIL-STD-810 (see 4.6.4.7) and shall meet the requirements stated therein.

3.3.2 Water resistance. The cured product shall be subjected to a water resistance test in accordance with method 512.3 of MIL-STD-810 (see 4.6.4.8) and shall meet the requirements stated therein.

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3.4 Storage life. The premixed and frozen compound, when stored in sealed containers, shall meet the requirements of 3.2 after a storage period of not less than 30 days, at -40°C or colder (see 4.7). Vendor certification in lieu of testing shall be acceptable (see 6.2).

3.5 Marking for identification. Marking for identification shall be in accordance with MIL-STD-130 (see 5.3).

3.6 Workmanship. The adhesive compound shall be prepared in a workmanlike manner and shall be free of apparent foreign particles (see 4.8).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When specified in the contract or purchase order (see 6.2) a first article sample shall be subjected to first article inspection. The quantity shall consist of one quart of adhesive

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manufactured by the same processes and using the same materials as the production article. First article inspection shall consist of all examinations and tests specified herein. Subsequent units shall not be considered for acceptance until Government approval of the first article sample has been obtained.

4.4 Quality conformance inspection. Quality conformance inspection for acceptance of the adhesive shall consist of the examinations and tests shown in table I.

TABLE I. Quality conformance inspection.

Examination and test	Requirement paragraph	Method paragraph
Overlap shear strength	3.2.2.1	4.6.4.1
Hardness, Shore A	3.2.2.2	4.6.4.2
Marking for identification	3.5	5.3
Workmanship	3.6	4.8
Packaging	Section 5	4.9

4.4.1 Lot size. Lot size shall consist of all the adhesive submitted for acceptance at the same time which has been prepared by the same company without a change in materials or processes in one continuous period of operation.

4.4.2 Sampling. Unless otherwise specified (see 6.2), one pint of adhesive from one unit, representative of each lot, shall be selected at random for quality conformance testing. Each container of adhesive shall be considered as a unit of product. Failure of the sample to meet the quality conformance requirements shall be cause for a lot rejection.

4.5 Inspection conditions. Unless otherwise specified, all inspections shall be performed under ambient conditions.

4.5.1 Inspection equipment accuracy. Equipment used to measure test conditions and test parameters shall be calibrated in accordance with MIL-STD-45662. Where the state of the art permits, basic precision equipment capable of measuring or testing to an accuracy representing one-tenth of the prescribed tolerance of the inspection equipment shall be used.

4.5.2 Test sequence. Testing sequence shall be in the order listed in table II.

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TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Specific gravity	3.2.1.1	4.6.2
Working life	3.2.1.2	4.6.3
Overlap shear strength	3.2.2.1	4.6.4.1
Hardness, Shore A	3.2.2.2	4.6.4.2
Dielectric constant	3.2.2.3	4.6.4.3
Volume resistivity	3.2.2.4	4.6.4.4
Copper corrosion	3.2.2.5	4.6.4.5
Fungus resistance	3.2.2.6	4.6.4.6
Hydrolytic stability	3.2.2.7	4.6.4.4
Storage life	3.4	4.7
Marking for identification	3.5	5.3
Workmanship	3.6	4, 8

4.6 Test methods.

4.6.1 Uncured properties.

4.6.2 Specific gravity. Specific gravity shall be determined in accordance with ASTM D792, method A (see 3.2.1.1).

4.6.3 Working life. Working life shall be determined by thawing out the compound in accordance with manufacturer's recommendations and applying the conditions of 3.2.1.2.

4.6.4 Cured properties.

4.6.4.1 Overlap shear strength. Overlap shear strength shall be determined in accordance with MMM-A-132 (see 3.2.2.1).

4.6.4.2 Hardness, Shore A. Shore A hardness shall be determined in accordance with ASTM D2240 (see 3.2.2.2).

4.6.4.3 Dielectric constant. Dielectric constant shall be determined in accordance with ASTM D150 (see 3.2.2.3).

4.6.4.4 Hydrolytic stability and volume resistivity. Vendor certification is acceptable in lieu of testing (see 3.2.2.7 and figure 1).

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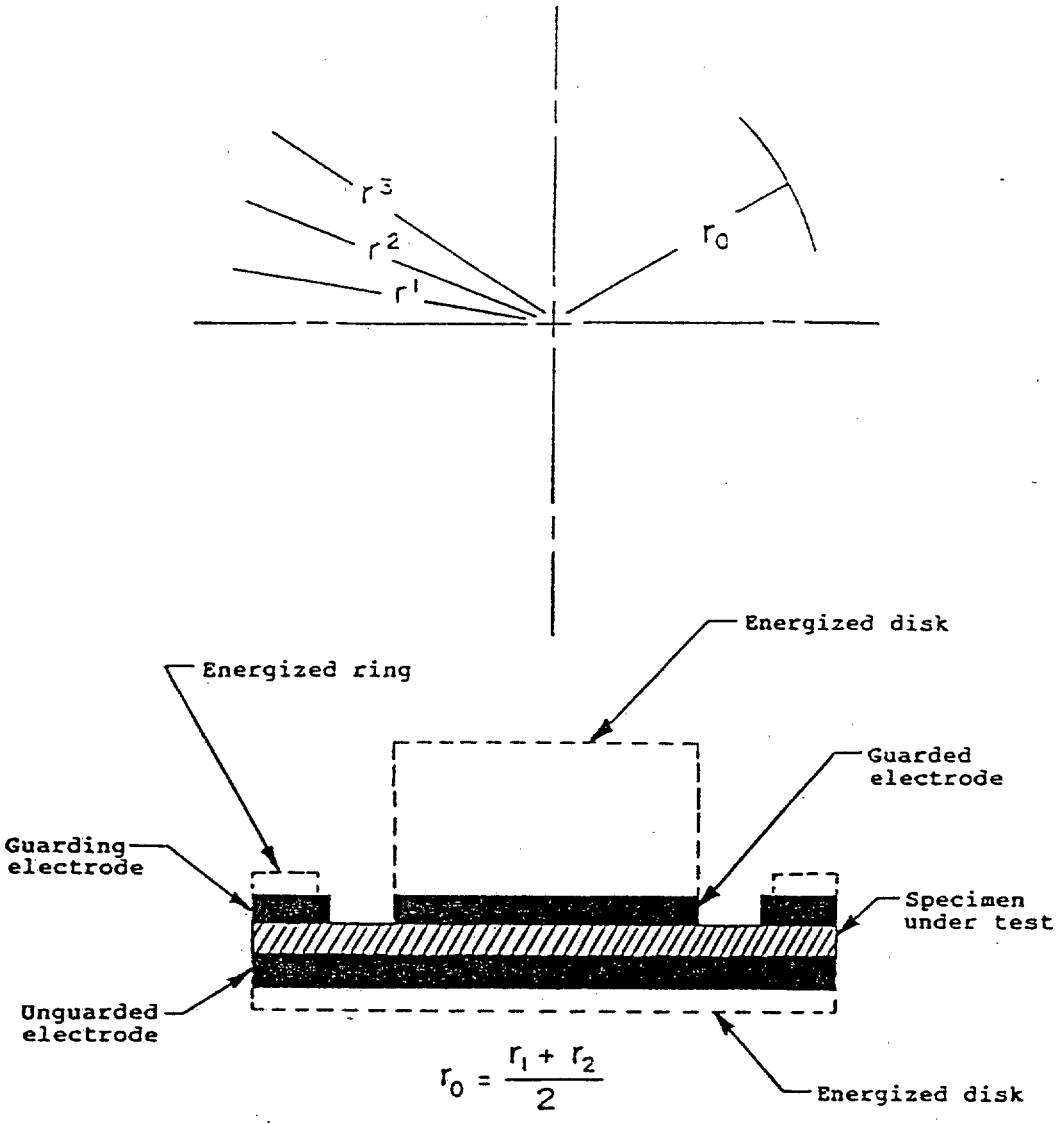


FIGURE 1. Volume resistivity specimen.

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4.6.4.4.1 Specimens. Four hardness specimens, approximately 2 inches in diameter and 0.250 inch thick, shall be prepared. Six specimens, 4 inches in diameter and approximately 0.075 inch thick, shall be prepared for volume resistivity tests (see 3.2.2.4).

4.6.4.4.2 Conditioning. Two hardness specimens and three volume resistivity specimens shall be maintained for 120 days at each of the following conditions:

25°C and 50% relative humidity
71°C and 95% relative humidity

4.6.4.4.3 Procedure. This test may be conducted in a chamber instead of a sealed apparatus. The specimens shall be tested for hardness and volume resistivity after 0, 28, 84 and 120 days. The 71°C specimens shall be held at 25°C and 50% relative humidity for 2 hours prior to testing. The instantaneous hardness shall be determined using a Shore A durometer, in accordance with ASTM D2240. Volume resistivity tests shall be conducted in accordance with ASTM D257, using a Beckman Megohmmeter Model L-8 or equivalent instrument with a test voltage of 500 volts. Readings shall be made one minute after application of current. Lead foil electrodes shall be a disk two inches in diameter, centrally located on one face of the specimen. The guard electrode shall be a concentric ring with an inside diameter of 2.281 inches and with an outside diameter equal to that of the specimen. The unguarded electrode shall be a foil disk 4 inches in diameter applied to the opposite side of the specimen. The test current shall be introduced to the guarded electrode, the guard electrode, and the unguarded electrode by means of a brass disk 2 inches in diameter by 1 inch in thickness, a brass ring with an inside diameter of 2.313 inches, an outside diameter of 4 inches, 0.125 inch thick, and a brass disk 4 inches in diameter, respectively (see figure 1).

4.6.4.4.4 Calculation. The average volume resistivity shall be calculated as follows:

$$r = \frac{RA \times 10^6}{t}$$

Where:

r = Volume resistivity in ohm-centimeters.

R = Volume resistance in megohms.

A = $(r_0)^2$, effective area of guarded electrode in square centimeters (see figure 1).

t = Average thickness of specimens in centimeters.

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4.6.4.5 Copper corrosion. The copper corrosion test shall be performed in accordance with FED-STD-175, method 4031-1 (see 3.2.2.5).

4.6.4.6 Fungus resistance. Fungus resistance shall be determined in accordance with MIL-STD-810, method 508. Vendor certification is acceptable in lieu of testing (see 3.2.2.6).

4.6.4.7 Humidity. The humidity level of the cured products shall be determined by MIL-STD-810, method 507.3 (see 3.3.1).

4.6.4.8 Water resistance. The cured product shall be determined by MIL-STD-810, method 512.3 (see 3.3.2).

4.7 Storage life. The compound shall meet the requirements of 3.4 when stored in unopened containers for not less than 30 days at -40°C or colder. Vendor certification is acceptable in lieu of testing (see 3.4 and 6.2).

4.8 Visual inspections. Visual inspection shall be in accordance with the requirements of 3.6.

4.9 Inspection of packaging. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of MIL-STD-1190.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with the specifications and standards indicated for levels A, B, or C.

5.1.1 Level A and level B. The compound shall be packaged in accordance with the overseas packaging requirements of PPP-C-2020.

5.1.2 Level C. The compound shall be packaged in unit capacity containers as specified (see 6.2) and shall conform to requirements of MIL-STD-1190.

5.2 Levels of packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The compound shall be packed in accordance with overseas packing requirements of PPP-C-2020.

5.2.2 Level B. The compound shall be packed in accordance with 5.2.1, except that shipping containers shall be a domestic type.

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5.2.3 Level C. Materials packaged as specified (see 6.2) shall be packed in containers which will ensure acceptance by common carrier at lowest rates and provide safe delivery at the destination.

5.3 Marking. Each container of adhesive shall be legibly and durably marked in accordance with MIL-STD-129. Marking information shall include, but shall not necessarily be limited to, the following:

a. Title, number and date of this specification

b. Lot or batch number

c. Mixing ratios

d. Necessary precautionary notices.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The adhesive covered by this specification is intended to bond metals and glass to themselves and to each other.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. Quantity required and container size (see 3.2.1)

d. When first article is required (see 3.1 and 4.3)

e. Certification of fungus resistance in lieu of testing (see 4.6.4.6)

f. Official levels of packaging and packing required (see 5.1 and 5.2)

g. Sampling plan if other than specified (see 4.4.2)

h. Certification of storage life, in lieu of testing, if permissible (see 3.4 and 4.7).

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6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Subject term (keyword) listing.

Bonding material
Glue
Plastic bond

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Preparing activity:
Army - MI

Project No. 8040 - 0500

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-A-47315A (MI)	910927

3. DOCUMENT TITLE
ADHESIVE, POLYURETHANE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER U.S. ARMY MISSILE COMMAND	(1) Commercial (205) 876-6980	(2) AUTOVON 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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